Batch File Record-Count Control

1. Aim

To write a COBOL program to compare the expected and observed record counts and display `MATCH` if they are equal; otherwise, display `MISMATCH` with the absolute difference.

2. Algorithm

1. Start the program.
2. Declare variables for expected count, observed count, and difference.
3. Read the expected record count.
4. Read the observed record count.
5. Compare both counts.
6. If both counts are equal, display `MATCH`.
7. Otherwise, calculate the absolute difference and display `MISMATCH` with the difference.
8. Stop the program.

3. Approach

1. Declare three numeric variables — `EXPECTED-COUNT`, `OBSERVED-COUNT`, and `COUNT-DIFF` — in the `WORKING-STORAGE SECTION` to store the expected count, observed count, and difference.

2. Use the `ACCEPT` statement to read the expected record count and observed record count from the input.

3. Use the `IF` statement to compare `EXPECTED-COUNT` and `OBSERVED-COUNT`.

4. If both counts are equal, use the `DISPLAY` statement to print `MATCH`.

5. If the counts are unequal, use a nested `IF-ELSE` statement to determine which count is greater.

6. Use the `COMPUTE` statement to subtract the smaller count from the larger count and store the absolute difference in `COUNT-DIFF`.

7. Use the `DISPLAY` statement to print `MISMATCH` along with the calculated difference.

8. Finally, use the `STOP RUN` statement to terminate the program.

4. Program

```
IDENTIFICATION DIVISION.
PROGRAM-ID. RECORD-CHECK.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 EXPECTED-COUNT PIC 9(9).
01 OBSERVED-COUNT PIC 9(9).
01 COUNT-DIFF     PIC 9(9).

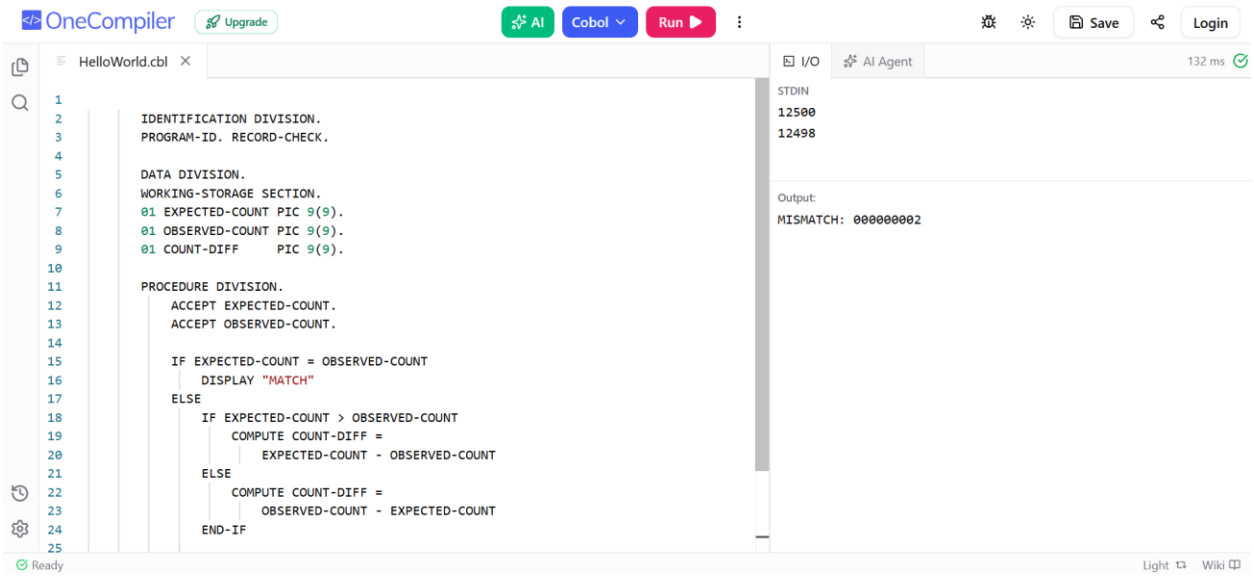
PROCEDURE DIVISION.
    ACCEPT EXPECTED-COUNT.
    ACCEPT OBSERVED-COUNT.

    IF EXPECTED-COUNT = OBSERVED-COUNT
        DISPLAY "MATCH"
    ELSE
        IF EXPECTED-COUNT > OBSERVED-COUNT
            COMPUTE COUNT-DIFF =
                EXPECTED-COUNT - OBSERVED-COUNT
        ELSE
            COMPUTE COUNT-DIFF =
                OBSERVED-COUNT - EXPECTED-COUNT
        END-IF

        DISPLAY "MISMATCH: " COUNT-DIFF
    END-IF.

    STOP RUN.
```

5. Output



6. Result

Thus, the COBOL program was successfully written and executed to compare the expected and observed record counts and display `MATCH` or `MISMATCH` with the absolute difference.